DATA SCIENCE × BUSINESS INTELLIGENCE

AI-driven mutual fund analytics

From transaction records to investor personas—and measurable marketing experiments.



CAPSTONE PROJECT

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ANALYTICS • SEGMENTATION • DECISION SUPPORT

The opportunity: turn activity into action

A connected workflow combines transaction analysis, behavioral segmentation and a decision-ready BI layer.

5,000
investor records
Reported project baseline

3
behavioral clusters
K-Means segmentation

6.56
transactions / investor
Reported average

₹352.16 Cr
gross turnover
All transaction types

THE ANALYTICAL JOURNEY

01 **Prepare**
Clean fields and scale features

02 **Segment**
Apply K-Means to behavior

03 **Explain**
Profile the clusters

04 **Activate**
Explore results in Power BI

Business value: move from retrospective reporting toward testable audience hypotheses—then measure whether campaigns improve outcomes.

Historical dashboards explain what happened

The next step is to make the analysis useful for decisions—without overstating what the data proves.

01

DATA VOLUME

Multiple transaction types and large monetary values make it easy to lose the story in totals.

02

BASIC RULES

Age or gender alone may miss meaningful differences in income, transaction frequency and invested volume.

03

STATIC REPORTING

A dashboard can describe activity, but audience hypotheses need a clear route to testing and measurement.

DESIGN QUESTION

How can we discover distinct investor behaviors and translate them into responsible, measurable marketing experiments?

CORRECTED TRANSACTION BASELINE

₹352.16 Cr gross turnover; ₹103.25 Cr net inflow

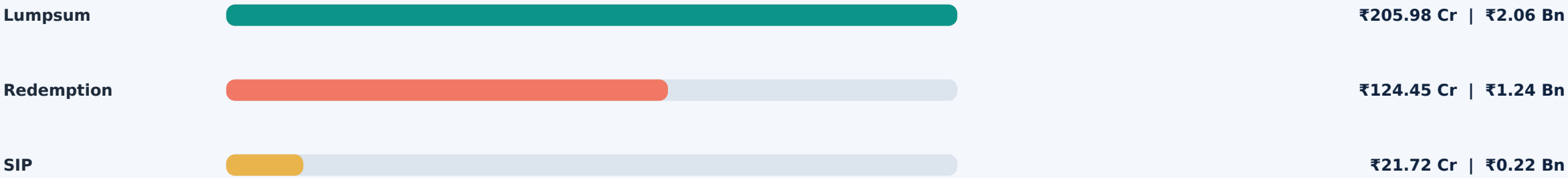
Reconciled from the applied transaction-level totals. Gross turnover equals gross inflow plus redemptions.

₹352.16 Cr	₹227.71 Cr	₹124.45 Cr	₹103.25 Cr
Gross turnover	Gross inflow	Redemptions / outflow	Net inflow

NET INFLOW

₹227.71 Cr gross inflow − ₹124.45 Cr redemptions = ₹103.25 Cr

COMPONENTS OF GROSS TURNOVER



KEY CORRECTION The old ~₹2.0 Bn figure maps to Lumpsum—not SIP. SIP volume is ₹0.22 Bn.

Lumpsum dominates the reported transaction mix

Share is calculated against total gross turnover of ₹3,521,580,430.

Lumpsum

₹2,059,821,448



₹205.98 Cr • ₹2.06 Bn

Redemption

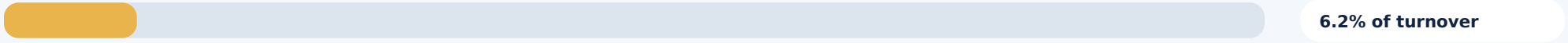
₹1,244,525,491



₹124.45 Cr • ₹1.24 Bn

SIP

₹217,233,491



₹21.72 Cr • ₹0.22 Bn

READOUT **Lumpsum is the largest category; SIP contributes 6.2% of gross turnover.**

CORE SOLUTION

A two-layer pipeline turns analysis into a usable product

Python prepares and segments the data; Power BI makes the outputs explorable.

01

INGEST

Source CSVs

Raw investor + transaction data

→

02

PREPARE

Pandas + scaling

Null handling • feature preparation

→

03

MODEL

K-Means

Cluster assignment + profiles

→

04

PUBLISH

Power BI

Measures • filters • dashboard

DATA SCIENCE LAYER

Pandas • scikit-learn • StandardScaler

Output: clustered_investors.csv

BUSINESS INTELLIGENCE LAYER

Power BI model • DAX measures • filters

Output: interactive decision-support dashboard

Implementation principle: keep transformations, cluster definitions and displayed metrics traceable so the dashboard can be audited and refreshed.

Give every number a clear business meaning

Use consistent labels, currency units and inflow/outflow semantics across cards, charts and tooltips.

GROSS TURNOVER

₹352.16 Cr

All three transaction types

A measure of transaction activity—not net money entering the fund.

GROSS INFLOW

₹227.71 Cr

Lumpsum + SIP

Show subscriptions separately from redemptions for a clear inflow view.

NET INFLOW

₹103.25 Cr

Gross inflow − redemptions

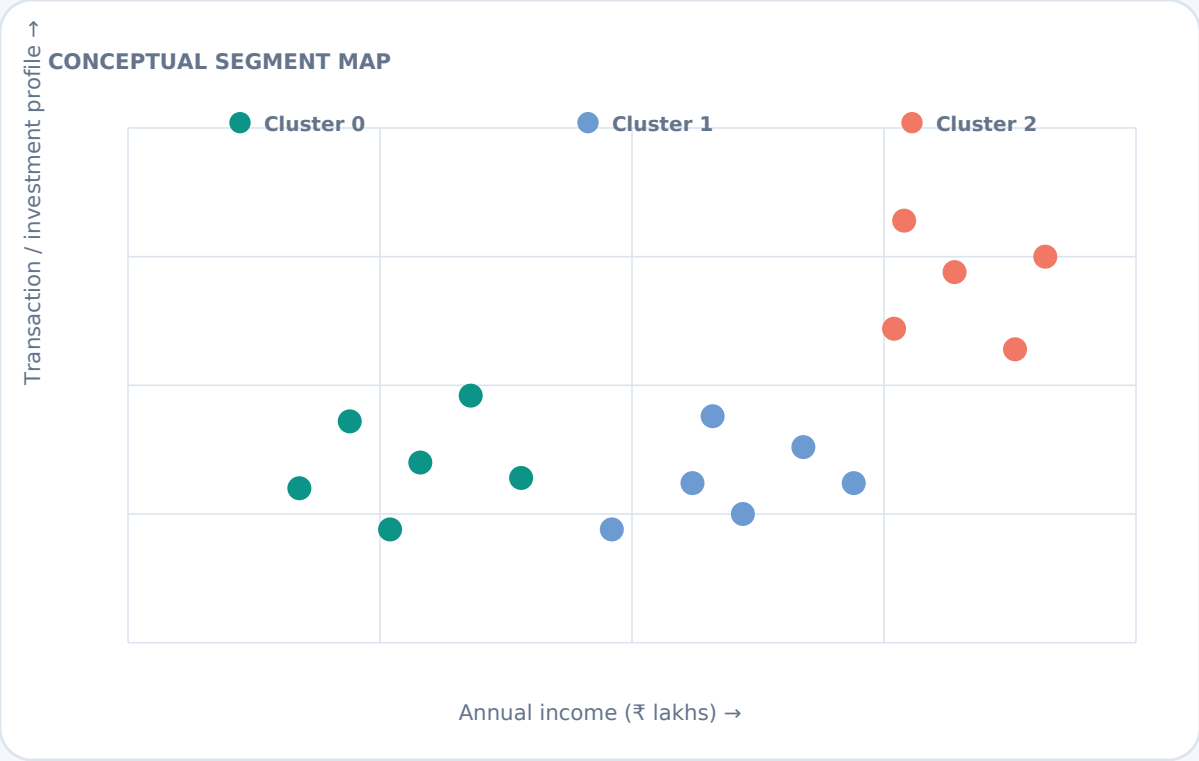
A net measure; do not confuse it with gross turnover.

DISPLAY STANDARD

- **Currency** — Use ₹ for rupee values; label Cr and Bn explicitly.
- **Direction** — Mark redemption as outflow; never mix it into gross inflow.
- **Precision** — Use consistent decimal places for headline values.
- **Traceability** — Tooltips should expose raw totals behind rounded values.

K-Means surfaces three reported behavioral groups

The project describes clusters using annual income, transaction frequency and invested volume.



WHAT THE MODEL USES

- **Annual income**
  Scale and distribution matter.
- **Transaction frequency**
  Distinguishes activity patterns.
- **Invested volume**
  Adds a capital-allocation view.

Illustrative points only—not raw model output. Validate cluster quality and stability before activation.

AI Persona Scorecard

Translate verified K-Means data profiles into targeted business audiences.

CLUSTER / VERIFIED PERSONA	INVESTORS	AVG. INCOME	ACTIONABLE STRATEGY
Cluster 0 **Moderate Retail** (Active Builders)	**1,958**	**₹19.6 Lakhs**	**Upsell diversified index funds**
Cluster 1 **Emerging Youth** (Entry Retail)	**2,147**	**₹17.3 Lakhs**	**Automate micro-SIPs & digital engagement**
Cluster 2 **Affluent HNI** (High Net Worth)	**895**	**₹62.4 Lakhs**	**Pitch Portfolio Management Services (PMS)**

SPEAKER NOTES

The supplied Python-output audit identifies Cluster 2 as the high-net-worth segment (average income above ₹62 Lakhs), while Cluster 1 is the entry-retail / emerging-youth segment. Separate wealth-management outreach from automated digital micro-investing.

Strategic Recommendations

Data-driven actions based on behavioral clustering.

01 AFFLUENT HNIs

Deploy PMS for Affluent HNIs

Cluster 2 • 895 investors

The supplied profile reports an average income of ₹62.4 Lakhs. Prioritise eligible investors for high-touch wealth-management conversations and tax-saving equity schemes, subject to suitability and compliance checks.

PMS / premium advisory

02 EMERGING YOUTH

Automate Emerging Youth

Cluster 1 • 2,147 investors

The supplied analysis states 70% are under 35 and transaction frequency is lower. Use automated micro-SIP nudges, digital onboarding and gamified education to encourage consistent investing.

Micro-SIPs / digital journeys

03 MODERATE RETAIL

Upsell Moderate Retail

Cluster 0 • 1,958 investors

The supplied analysis describes this group as averaging nearly 10 transactions each. Test diversified index-fund education and broader asset-allocation journeys with clear risk disclosures.

Index funds / asset allocation

SPEAKER NOTES

Because the segmentation is derived from observed transaction behavior, use it to formulate differentiated campaign hypotheses—not to assume individual preferences. Validate the profiles, confirm eligibility and suitability, then run controlled pilots and measure outcomes against a baseline.

Protect the analysis from data and interpretation risk

Good presentation design is important—but metric integrity and model validation come first.

OUTLIER SKEW

RISK Ultra-high income values can pull K-Means centroids away from typical investor patterns.

CONTROL Scale features, inspect distributions and compare cluster stability.

CURRENCY / UNIT CONFUSION

RISK Mixing ₹, lakh, crore and billion scales makes dashboard values easy to misread.

CONTROL Use explicit units, consistent precision and source-to-visual reconciliation.

LABEL OVERREACH

RISK Cluster names can sound like confirmed investor identities when they are only interpretations.

CONTROL Treat personas as hypotheses; validate with domain review and controlled pilots.

TARGETING FAIRNESS

RISK Segment-based outreach can create privacy or fairness concerns if used without controls.

CONTROL Review data provenance, privacy permissions and campaign eligibility.

From insight to action.

Validate the personas. Pilot a campaign. Measure the result against a baseline.

ILLUSTRATIVE TARGETS TO VALIDATE

+15%
projected SIP retention lift
Target—not a measured outcome

12 hrs
projected monthly time saved
Target—not a measured outcome

3×
projected targeting precision
Target—not a measured outcome

RECOMMENDED NEXT STEPS

1. Validate segment definitions
2. Pilot a targeted campaign
3. Measure vs. baseline
4. Iterate from evidence

SOURCE CODE

github.com/tanay1704/bluestock-fintech

The goal is not simply to build a dashboard—it is to create a repeatable, measurable decision process.

Tanay Jain • Data Science & Business Intelligence